

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Probability Diagrams - Venn & Tree Diagrams

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PROBABILITY DIAGRAMMS - VENN & TREE DIAGRAMMS · CH6 · L19

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The Map

This lesson collapses Probability Diagrams - Venn & Tree Diagrams into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

02 Use complement

TRIGGER *"probability", "random", "given that"*

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 29 website questions, 29 checked solutions, 3 local PDFs read.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(A \text{ odd}) = \frac{7}{10}, \quad P(A \text{ even}) = \frac{3}{10}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q16

Calculate probability

$$P(A \text{ odd}) = \frac{7}{10}, \quad P(A \text{ even}) = \frac{3}{10}$$

Calculate probability

$$\frac{7}{10} \times \frac{5}{12} + \frac{3}{10} \times \frac{7}{12} = \frac{7}{15}$$

Finish: $\frac{7}{15}$, and 25.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

02 Use complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(A \text{ odd}) = \frac{7}{10}, \quad P(A \text{ even}) = \frac{3}{10}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q16

Calculate probability

$$P(A \text{ odd}) = \frac{7}{10}, \quad P(A \text{ even}) = \frac{3}{10}$$

Calculate probability

$$\frac{7}{10} \times \frac{5}{12} + \frac{3}{10} \times \frac{7}{12} = \frac{7}{15}$$

Finish: $\frac{7}{15}$, and 25.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$0.6 \times 0.9 = 0.54$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q14

Winning both frames

$$0.6 \times 0.9 = 0.54$$

Winning exactly one frame

$$0.6 \times 0.1 + 0.4 \times 0.25$$

Finish: 0.54, 0.16.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q14

Calculate probability

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

Finish: $\frac{21}{190}$.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q14

Calculate probability

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

Finish: $\frac{21}{190}$.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"probability", "random", "given that"	Count wanted over total
"probability", "random", "given that"	Use complement
"probability", "random", "given that"	Multiply along tree branches
"probability", "random", "given that"	Use conditional probability
"probability", "random", "given that"	Calculate probability

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.